Optiguard

1st Kuro Narendra Maheswara
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

2nd Jeljel Wirayudha Pratama Belanegara
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

3rd Chiko Bagaskara Adiwijaya
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

4th Michi
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

5th Rocky
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

6th Piko
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

***Abstract*—Early and accurate detection of eye diseases is critical to preventing irreversible vision loss, yet most deep learning diagnostic models remain limited by their "black-box" nature and lack of quantitative interpretability validation. This study presents a comprehensive comparative evaluation between modern Convolutional Neural Networks (ConvNeXt-Tiny) and Vision Transformers (DeiT-Small) for multiclass retinal disease classification using fundus images (categorized into cataract, diabetic retinopathy, glaucoma, and normal classes). Beyond standard classification performance metrics—where DeiT-Small and ConvNeXt-Tiny achieved comparable overall macro-F1 scores of 87.74% and 87.63%, respectively—this work provides a rigorous quantitative assessment of visual explanation faithfulness. Specifically, we evaluate the explanation fidelity of Grad-CAM (for ConvNeXt) and Attention Rollout (for DeiT) using patch-based deletion and insertion Area Under the Curve (AUC) metrics. Our findings demonstrate that while both architectures exhibit strong discriminative power and clinical alignment on localized lesions (such as diabetic retinopathy and glaucoma), their explainability profiles differ significantly in spatial focus and faithfulness. This research advances medical image analysis by bridging the gap between classification accuracy and explainability validation, offering empirical guidelines for selecting optimal model-XAI combinations in clinical decision-support systems.**

***Index Terms*—VLM, surveillance, LoRA, DoRA**

I. Introduction

Vision impairment affects over 2.2 billion people globally. Major causes include cataracts, diabetic retinopathy, and glaucoma [1]. Cataracts account for 45% of global blindness (15.2 million people) and cause moderate-to-severe visual impairment in 78.8 million others [2]. Diabetic retinopathy affects 20%–35% of diabetic patients, with projected cases reaching 129.84 million by 2030 and 160.50 million by 2045 [3]. Meanwhile, glaucoma, as the leading cause of irreversible blindness, is projected to affect 111.8 million people by 2040, primarily in Asia and Africa [4]. These data emphasize the urgent need for scalable early detection systems, especially in low- and middle-income regions with limited eye care access.

To address this challenge, deep learning that particularly Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) has become foundational in medical image analysis by automatically extracting features from ophthalmological modalities without manual engineering [5]. Modern architectures like ConvNeXt modernize CNNs using Transformer design principles to enhance accuracy while preserving efficiency [6]. Conversely, DeiT advances ViTs by leveraging knowledge distillation to enable effective training on smaller datasets with reduced computational overhead [7].

Although deep learning models have very high classification accuracy, they often suffer from a black-box limitation. This means that the model can provide accurate predictions, but the process behind its decision-making is not transparent and is difficult for humans to understand [8]. Therefore, Explainable

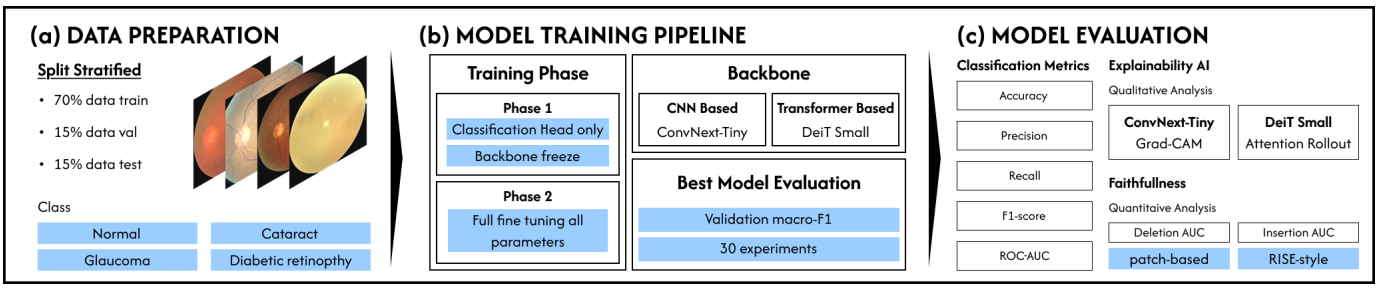


Fig. 1. Overview of the research methodology.

Artificial Intelligence (XAI) technology is used to visualize the reasoning behind the model's predictions. For models that use CNN architecture such as ConvNeXt, the Grad-CAM method is typically used to create a heatmap that highlights the most important areas of the image. Meanwhile, in Transformer architectures such as DeiT, the method used is Attention Rollout or Gradient Attention Rollout. This method works by collecting the attention weights of each layer of the model to see which parts of the image the model is most focused on [9]. The different internal mechanisms of these two models mean that the quality of the resulting visualizations is not directly comparable. The main problem is that current deep learning evaluations in the medical field mostly stop at accuracy comparison. To quantitatively measure how faithful these visualizations are, specific metrics are required. Deletion and insertion metrics that use Area Under Curve (AUC) values are often used as the standard of measurement. These metrics work by gradually hiding or showing parts of the image according to their importance, and then seeing how much the model's confidence level in its predictions changes [8].

Based on the problems that have been described, the focus of this research is formulated into two main analytical scopes. The first focus is on a comparative evaluation of the classification performance between the ConvNeXt and DeiT architectures on a dataset of four classes of eye diseases (cataracts, diabetic retinopathy, glaucoma, and normal) using accuracy, precision, recall, and F1-score metrics. The second focus is a comparative analysis of the quality of visual explanations, particularly at the level of faithfulness of heatmap representation, between the Grad-CAM method in the ConvNeXt model and the Attention Rollout in the DeiT model, which is quantitatively evaluated through deletion AUC and insertion AUC measurements.

To address these problems, this research presents two main contributions. Theoretically, this study enriches the academic literature related to the comparative interpretability between modern CNN (ConvNeXt) and Vision Transformer (DeiT) architectures that are specialized in the domain of ophthalmological medical imaging. previous research that rarely explored the comparison of models in depth based on the faithfulness parameters of XAI visualizations, instead of relying solely on accuracy metrics. Meanwhile, practically, this study provides an empirical evidence-based guideline for clinical practitioners and engineers in determining the combination of XAI architecture and methods that best suits the needs of operational systems. For example, the results of this research can be used as a direct reference in choosing the most recommended architecture when a diagnostic system requires a heatmap localization that is highly aligned with the model's decision-making basis.

II. Related Work

A. Deep Learning-Based Eye Disease Classification

Deep learning-based eye disease classification using fundus images has attracted significant research attention in recent years. The Eye Disease Classification dataset [10] is one of the publicly available datasets frequently used for benchmarking multiclass eye disease classification models. The characteristics of the dataset are presented in Table I.

TABLE I
Distribution of Eye Disease Classification Dataset

Class	Number of Images	Percentage (%)
Cataract	1,038	24.62
Diabetic Retinopathy	1,098	26.04
Glaucoma	1,007	23.88
Normal	1,074	25.47
Total	**4,217**	**100.00**

Dataset characteristics are relatively consistent across studies, comprising 4,217 fundus images categorized into four classes: cataract, diabetic retinopathy, glaucoma, and normal eyes [11], [12]. Although the dataset is relatively balanced, glaucoma represents the minority class, which motivates the use of data augmentation techniques during preprocessing. Furthermore, the dataset provides inherent inter-subject variability across different patients. Similar cross-subject challenges in biomedical data were highlighted by Colot et al. [13], emphasizing the need for robust feature evaluation beyond standard accuracy.

The dataset has been utilized in several studies to evaluate the performance of deep learning models. A comparative study involving DenseNet, EfficientNet, Xception, VGG, and ResNet was conducted in [11]. Among these architectures, EfficientNet achieved the highest accuracy of 87.84%. Subsequently, transfer learning with EfficientNet-B3 combined with data augmentation and cosine annealing learning rate scheduling was proposed in [14], resulting in an improved accuracy of 95.12%.

A transfer learning framework named TL-MED was introduced in [12], where five CNN-based architectures and an ensemble model were evaluated. DenseNet169 achieved the highest individual accuracy of 96%. In another study, AlexNet, EfficientNet-B0, EfficientNet-B7, and a custom CNN architecture were compared [15]. AlexNet demonstrated the best performance with an accuracy of 93.65%, and the study further employed SHAP-based interpretability analysis to identify clinically relevant image regions. Beyond SHAP, gradient-based visualization techniques such as Grad-CAM [16] have also been widely adopted in fundus image classification studies to highlight discriminative regions contributing to the model's prediction, providing a complementary means of interpreting CNN-based decisions.

Based on these studies, existing research on this dataset primarily focuses on conventional CNN architectures and their variants, including VGG, ResNet, DenseNet, EfficientNet, and AlexNet. In contrast, investigations involving modern CNN architectures such as ConvNeXt and transformer-based architectures such as DeiT remain limited. This research gap motivates the comparative analysis conducted in this study.

B. ConvNeXt and DeiT Architectures for Image Analysis

As representative architectures of modern convolutional networks and vision transformers, ConvNeXt and DeiT have demonstrated promising performance across various image classification tasks. ConvNeXt was introduced in [6] as a modernization of the ResNet architecture by incorporating several design principles inspired by Vision Transformers, including patchify stems, depthwise convolutions, inverted bottlenecks, larger kernel sizes, and simplified normalization and activation layers.

On the other hand, DeiT (Data-efficient Image Transformer) was proposed in [7] as an extension of the original Vision Transformer architecture [17]. Since the attention mechanism on its own does not take the order of tokens into consideration, positional encodings are integrated into the patch embeddings [18]. DeiT introduces a knowledge distillation strategy that enables transformer models to be trained effectively on relatively smaller datasets while maintaining competitive performance. Beyond classification, the adaptation of transformer architectures and attention mechanisms has also demonstrated strong efficacy across diverse computer vision domain tasks [19].

Several comparative studies have investigated the strengths and limitations of CNN and Vision Transformer architectures in image analysis tasks. A literature review presented in [20] concluded that CNNs benefit from strong inductive biases and superior data efficiency, whereas Vision Transformers are more effective at capturing global contextual information but generally require larger datasets and greater computational resources.

In the specific domain of fundus image analysis, a comparative study between CNN-based architectures, including ConvNeXt, and transformer-based models was conducted in [21]. The results showed that the transformer-based CaiT architecture achieved the highest AUC of 0.979, although the performance gap between the best CNN and transformer models was relatively small. Similar findings were reported in [22], where Vision Transformer models achieved non-inferior or superior performance compared with CNNs on five out of six glaucoma datasets.

Overall, these findings suggest that the relative advantages of modern CNN architectures such as ConvNeXt and transformer-based architectures such as DeiT depend heavily on dataset characteristics and training strategies. Therefore, no clear consensus has emerged regarding the superiority of either approach. This motivates a direct comparison between ConvNeXt and DeiT for multiclass eye disease classification using the Eye Disease Classification dataset.

III. Methodology

A. Model Architecture

1) ConvNeXt: ConvNeXt is a modern convolutional neural network (CNN) architecture inspired by Vision Transformers (ViTs). It modernizes standard convolutional operations without relying on self-attention mechanisms. Specifically, ConvNeXt redesigns the standard ResNet architecture by adopting 7×7 depthwise convolution to enlarge the receptive field, using inverted bottleneck blocks, replacing Batch Normalization with LayerNorm, and replacing Rectified Linear Unit (ReLU) with GELU to improve optimization stability during transfer learning [6]. These architectural refinements enhance feature representation without sacrificing the computational advantages of convolution-based networks. Fig. 2 illustrates the internal structural flow of a ConvNeXt block.

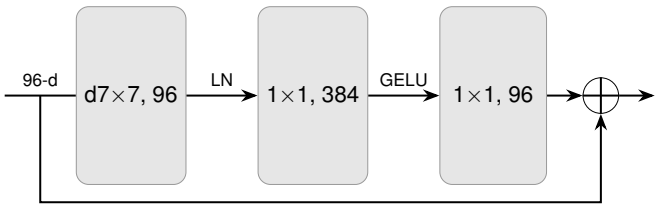


Fig. 2. ConvNeXt Block Architecture.

To evaluate its real-world efficacy, a recent large-scale study compared ConvNeXt, RegNet, and ResNet across 22 publicly available retinal fundus datasets [23]. This benchmark comprised more than $125,000$ images spanning normal cases, diabetic retinopathy, glaucoma, and age-related macular degeneration to construct a clinically representative screening model. Among the tested architectures, ConvNeXt-Tiny achieved the best overall performance across most evaluation metrics, obtaining an overall accuracy of $80.46 \pm 1.48\%$. It demonstrated particularly high disease recognition accuracy for glaucoma (97.20%), age-related macular degeneration (98.14%), and diabetic retinopathy (80.66%) These findings demonstrated ConvNeXt-Tiny's reliability for retinal disease classification while maintaining a relatively lightweight architecture, making it highly suitable for transfer learning on moderately sized retinal fundus datasets.

2) DeiT: Standard Vision Transformers (ViTs) rely primarily on self-attention mechanisms to capture long-range dependencies across an entire image and generally require extremely large-scale pretraining datasets to achieve competitive performance [17]. To overcome these substantial data requirements while preserving the advantages of transformer-based architectures, Data-efficient Image Transformers (DeiT) were proposed [7]. DeiT introduces a dedicated distillation token into the architecture, which is appended alongside the class token and patch embeddings as illustrated in Fig. 3. During training, this token interacts with the patch tokens via self-attention to learn complementary features from a robust, pretrained teacher model (typically a CNN). This mechanism significantly improves classification accuracy, particularly for smaller transformer models, while maintaining computational efficiency.

The data-efficient design of DeiT makes it particularly suitable for medical image analysis, where large annotated datasets are often unavailable and training standard ViTs from scratch is difficult. A recent study incorporated DeiT as one of the transformer-based feature extractors in a hybrid framework for multi-class eye disease classification, demonstrating that

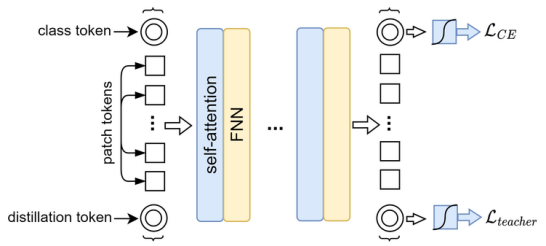


Fig. 3. DeiT Block Architecture

combining DeiT with machine learning classifiers contributed to a stacking model that achieved 97.163% classification accuracy on a benchmark retinal dataset [24]. These findings indicate that transformer-based models can be effectively applied to retinal disease classification when equipped with efficient training or feature extraction strategies. Therefore, DeiT is suitable for transfer learning as a representative data-efficient Vision Transformer for a moderately sized retinal fundus datasets.

3) Hyperparameter & Model Selection: Hyperparameters are user-defined settings that control the training process, whereas model parameters are the network weights automatically learned through backpropagation [25]. Both ConvNeXt and DeiT were fine-tuned using the AdamW optimizer [26] with a cosine learning-rate decay schedule, following the official training recipes of both architectures. Although the original papers employ different optimization settings, particularly for large-scale ImageNet pretraining, reproducing their exact recipes was impractical given the hardware constraints and the smaller size of the diabetic retinopathy dataset. A unified configuration was therefore adopted for both models to ensure a fair comparison.

Rather than exhaustively tuning every hyperparameter, only the fine-tuning learning rate and the fine-tuning epoch budget were optimized, using a grid search over three learning rates (10^{-4}, 5×10^{-5}, and 10^{-5}) and five epoch budgets (10, 20, 30, 40, and 50), giving 15 configurations per architecture. Restricting the search to these two hyperparameters follows Bergstra and Bengio [27], who showed that only a small subset of hyperparameters typically has a significant impact on model performance, which keeps an exhaustive grid over the remaining search space computationally tractable. All remaining settings were fixed across both architectures and selected from their official training recipes or adjusted to the available computational budget. For each architecture, the configuration with the highest validation macro-F1 was selected. Table II summarizes the final configuration, where values that differ between the two models are given as ConvNeXt / DeiT.

To ensure an unbiased evaluation, the dataset is divided into training, validation, and testing subsets. the training set optimized the model parameters, the validation set selected the best hyperparameter configuration [28], and the test set was reserved solely for final performance evaluation, without involvement in any stage of model selection [29]. This separation prevents information leakage and optimistic bias, yielding a more reliable estimate of the model's generalization capability.

TABLE II
FINAL HYPERPARAMETER CONFIGURATION.

Hyperparameter	**Value**	**Source / Justification**
Optimizer	AdamW	Common optimizer for both models
LR scheduler	Cosine decay	Adopted from both models' official recipes
Weight decay	0.05	Adopted from DeiT training recipe and applied consistently to both models
Label smoothing	0.1	Recommended by both ConvNeXt and DeiT papers
Linear probe LR	10^{-3}	Empirically selected for the linear probing stage
Linear probe epochs	5	Chosen to initialize the classifier before fine-tuning
Fine-tuning epochs	30 / 40	Grid search over $\{10, 20, 30, 40, 50\}$
Fine-tuning LR	10^{-5} / 10^{-4}	Grid search over $\{10^{-4}, 5 \times 10^{-5}, 10^{-5}\}$
Batch size	32	Limited by available GPU memory

B. Explainable AI (XAI) on Computer Vision

1) Grad-CAM: For a target class c, Grad-CAM first back-propagates the class score y^c with respect to each feature map A^k in the final convolutional layer. The importance weight of the k-th feature map is obtained by applying global average pooling to the spatial gradients, which is defined as

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{i,j}^k} \tag{1}$$

where Z denotes the total number of spatial locations in the feature map. The resulting weight α_k^c represents the contribution of feature map A^k to the prediction of class c [30].

The class activation map is subsequently generated as a weighted combination of all feature maps,

$$L_{Grad-CAM}^c = ReLU(\sum_k \alpha_k^c A^k) \tag{2}$$

where the ReLU suppresses negative activations, preserving only features that positively influence the prediction of the target class [30].

Although originally developed for CNNs, Grad-CAM has been successfully adapted to Vision Transformers. The method utilizes the patch token representations extracted from the final transformer encoder block, which are reshaped into a two-dimensional spatial grid before gradient computation. The spatial gradients are globally averaged to obtain token importance weights, followed by weighted aggregation and ReLU activation to produce the final saliency map [31], [32].

Grad-CAM has been widely adopted in medical image analysis to verify whether deep learning models focus on clinically relevant anatomical structures or pathological lesions. As summarized in Table III, Grad-CAM has demonstrated its effectiveness across multiple imaging modalities and deep learning architectures.

TABLE III
APPLICATIONS OF GRAD-CAM IN MEDICAL IMAGE ANALYSIS.

Medical Domain	Architecture	Interpretability Outcome
Chest X-ray (TB)	ViT	Highlights diagnostically relevant pulmonary regions, enabling visualization of the model's focus during tuberculosis classification [32].
Retinal disease	ResNet101-V2	Emphasizes retinal abnormalities associated with glaucoma, diabetic retinopathy, and cataracts, improving multiclass model interpretability [33].
Thyroid ultrasound	Hybrid CNN–ViT	Produces clinically meaningful activation patterns that validate lesion localization and support prediction reliability [31].

2) Attention Rollout: Vision Transformers compute a self-attention matrix at every transformer layer, where each element represents the attention weight assigned from one token to another. Since each image patch is represented as a token, the attention matrix describes how information is exchanged among all image patches through the self-attention mechanism. Self-attention enables every token to interact with all other tokens, allowing the model to capture global contextual relationships, unlike convolutional neural networks that rely on local receptive fields [17].

However, the attention matrix from a single transformer layer only reflects the interaction at that particular layer and does not directly indicate the overall contribution of an input patch to the final prediction. To estimate how information propagates throughout the transformer, Attention Rollout recursively composes the attention matrices from consecutive layers. Consequently, the importance of each input patch is determined by the accumulated attention flow from the input layer to the final class token representation [34].

The recursive formulation of Attention Rollout is expressed as

$$\tilde{A}(l_i) = \begin{cases} A(l_i)\tilde{A}(l_{i-1}) & \text{if } i > j \\ A(l_i) & \text{if } i = j \end{cases} \tag{3}$$

where $A(l_i)$ denotes the attention matrix of the i-th transformer layer, while $\tilde{A}(l_i)$ represents the accumulated attention after recursively composing the attention matrices from previous layers. For image classification, the attention values corresponding to the class (CLS) token in the final rollout matrix are reshaped into the spatial arrangement of the input patches to generate the explanation heatmap [34].

Compared with gradient-based explanation methods, Attention Rollout provides a global interpretation by tracing information flow through all transformer layers rather than relying on the gradients of a single layer. The resulting heatmaps tend to be more diffused and less spatially localized than those produced by Grad-CAM, because the explanation aggregates attention across multiple layers [35], [36]. A summary of the differences between Attention Rollout and Grad-CAM is presented in Table IV.

TABLE IV
COMPARISON BETWEEN GRAD-CAM AND ATTENTION ROLLOUT

Method	Characteristics
Grad-CAM	Produces class-discriminative heatmaps using gradients from the last convolutional feature map. It provides relatively sharp and localized visual explanations but is primarily designed for CNNs.
Attention Rollout	Propagates attention recursively across transformer layers to estimate the contribution of each input patch to the CLS token. It captures global contextual relationships but generally produces more diffuse heatmaps than Grad-CAM.

C. Faithfulness Evaluation

The quality of the generated saliency maps was quantitatively evaluated using the deletion and insertion Area Under the Curve (AUC) metrics. These metrics are both model-agnostic and explanation-agnostic because they assess explanation quality based solely on changes in the model prediction after perturbing the input image, regardless of the internal mechanism used to generate the explanation [37], [38]. Deletion AUC measures whether the regions identified as important by a saliency map are truly critical for the model prediction, while insertion AUC measures how effectively the highlighted regions restore the model prediction.

The original formulation perturbs the image pixel by pixel [37]. In this study, the perturbation is instead applied at the patch level: every heatmap is expressed on a common 14×14 grid of 16×16-pixel patches over the 224×224 input, which is the native patch grid of DeiT, and the Grad-CAM map of ConvNeXt is bilinearly resized to the same grid. Using the same perturbation granularity for both architectures keeps their faithfulness scores directly comparable. As the uninformative baseline, a heavily Gaussian-blurred copy of the input image (kernel size 51, $\sigma = 41$) is used rather than a constant black or gray image, so that perturbed regions do not introduce sharp artificial edges that the model has never seen during training.

Deletion AUC starts by ranking all 196 patches according to their importance values in the generated heatmap. The most important patches are then progressively replaced, one patch per step, with the corresponding region of the blurred baseline while the softmax confidence of the predicted class is recorded after each step. The AUC is then calculated with the trapezoidal rule from the resulting confidence values plotted against the fraction of removed patches, where a lower AUC indicates a more faithful explanation because the model

confidence decreases rapidly when the truly relevant regions are removed [35].

Insertion AUC starts from the blurred baseline image, and the original patches are gradually inserted in the same order of importance. The AUC is then calculated from the prediction confidence plotted against the fraction of inserted patches, where a higher AUC indicates a more faithful explanation because the selected patches rapidly recover the confidence of the target prediction [35].

IV. Results

This section presents the experimental results of the study, organized according to the two analytical scopes defined in Section I: (1) the classification performance comparison between ConvNeXt-Tiny and DeiT-Small, and (2) the faithfulness comparison between Grad-CAM and Attention Rollout as measured by deletion and insertion AUC.

A. Hyperparameter Selection

Both ConvNeXt-Tiny and DeiT-Small were fine-tuned using a two-phase training scheme (linear-probe followed by full fine-tuning) under a grid search over learning rate and number of training epochs. For each architecture, the configuration achieving the highest validation macro-F1 was selected as the final model, following the model selection protocol described in Section III-A. Table V summarizes the selected configuration for each architecture.

TABLE V
Selected Hyperparameter Configuration per Architecture

Hyperparameter	ConvNeXt-Tiny	DeiT-Small
Linear probe learning rate	1×10^{-3}	1×10^{-3}
Full fine-tuning learning rate	1×10^{-5}	1×10^{-4}
Weight decay	0.05	0.05
Label smoothing	0.1	0.1
Linear probe epochs	5	5
Full fine-tuning epoch budget	30	40
Batch size	32	32

The selected configuration for each architecture converged well before its configured epoch budget was exhausted (epoch 11 of 30 for ConvNeXt-Tiny, epoch 19 of 40 for DeiT-Small), consistent with the two-phase fine-tuning scheme reaching its best validation macro-F1 early in the full-fine-tuning phase. The full ranking of all 15 grid-search combinations per architecture is available in the accompanying evaluation notebook (`eval.ipynb`).

B. Classification Performance

Table VI presents the test-set classification performance of both architectures, reported as overall accuracy and macro-averaged precision, recall, and F1-score. Diabetic retinopathy recall is reported separately, as it serves as the tie-breaker criterion for final model selection due to the higher clinical cost of false-negative diagnoses.

Table VII further reports the per-class precision, recall, and F1-score for the winning model, providing a more detailed view of class-wise classification behavior, particularly for the minority glaucoma class.

TABLE VI
Test-Set Classification Performance

Metric	ConvNeXt-Tiny	DeiT-Small
Accuracy	87.99%	87.99%
Macro Precision	87.91%	88.03%
Macro Recall	87.74%	87.77%
Macro F1-score	87.63%	**87.74%**
Macro ROC-AUC	**98.15%**	97.06%
DR Recall (tie-breaker)	**99.39%**	95.76%

TABLE VII
Per-Class Performance of the Selected Model (DeiT-Small)

Class	Precision	Recall	F1-score
Cataract	93.46%	91.67%	92.56%
Diabetic Retinopathy	87.78%	95.76%	91.59%
Glaucoma	86.26%	74.83%	80.14%
Normal	84.62%	88.82%	86.67%

Based on the test-set macro-F1, DeiT-Small (87.74%) marginally outperformed ConvNeXt-Tiny (87.63%) – a gap of only 0.11 percentage points, small enough that the two architectures are effectively comparable in overall discriminative power. Since the macro-F1 values were not exactly tied, the diabetic retinopathy recall tie-breaker was not formally invoked; however, it is worth noting that ConvNeXt-Tiny attained a substantially higher DR recall (99.39% vs. 95.76%) and macro ROC-AUC (98.15% vs. 97.06%), a discrepancy discussed further in Section IV-E. Per the strict macro-F1 ranking, DeiT-Small is treated as the best-performing architecture and is used as the subject of the per-class analysis above; both architectures' checkpoints are nonetheless carried forward into the explainability analysis in the following subsections, since faithfulness is compared architecture-to-architecture (Grad-CAM vs. Attention Rollout) rather than reduced to a single winner. Both models show the same qualitative weak point: glaucoma has the lowest recall of the four classes (74.83% for DeiT-Small), reflecting its status as the dataset's minority class (Table I).

Fig. 4 shows the row-normalized confusion matrix of the selected model (DeiT-Small) on the test set, confirming that glaucoma is most frequently confused with cataract and normal, while diabetic retinopathy and cataract are both classified with high recall.

C. Qualitative Explainability Results

Fig. 5 presents qualitative examples of Grad-CAM heatmaps generated from ConvNeXt-Tiny and Attention Rollout heatmaps generated from DeiT-Small on representative test images from each class. For glaucoma and diabetic retinopathy, both methods concentrate their highest activations tightly on the optic disc region – the clinically relevant structure for assessing cupping in glaucoma and a common site of

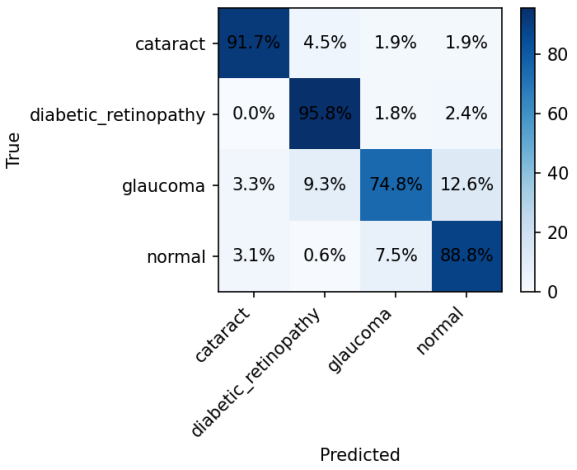


Fig. 4. Normalized confusion matrix of the selected model (DeiT-Small) on the test set.

hemorrhages and exudates in diabetic retinopathy – indicating that both models learned to attend to anatomically meaningful evidence rather than spurious background cues for these two classes. For the normal example, both methods likewise center on the optic disc and surrounding vasculature, consistent with this being the reference structure clinicians inspect first. Cataract is the weakest case for both methods: Grad-CAM activates a narrow vertical strip near the image border rather than a diffuse haze pattern (the actual visual signature of a cataract in a fundus photo), while Attention Rollout spreads its attention toward the image corners and edges outside the fundus circle. This is consistent with cataract's presentation as a global loss of image sharpness/contrast rather than a spatially localized lesion, which is inherently harder for patch-level saliency methods – both CNN- and attention-based – to localize. Attention Rollout's tendency to assign non-trivial weight to border/corner patches across multiple classes also matches its characterization in Section III-B as producing more diffuse heatmaps than Grad-CAM.

D. Faithfulness Evaluation

To quantitatively assess the faithfulness of the generated explanations, deletion and insertion AUC were computed for Grad-CAM on ConvNeXt-Tiny and Attention Rollout on DeiT-Small using patch-based perturbation, as described in Section III-C. Table VIII summarizes the resulting scores, averaged across the test set.

TABLE VIII
DELETION AND INSERTION AUC OF GRAD-CAM AND ATTENTION ROLLOUT ($n = 40$ SAMPLED TEST IMAGES)

XAI Method	Deletion AUC ↓	Insertion AUC ↑
Grad-CAM (ConvNeXt-Tiny)	**0.3906**	0.6982
Attention Rollout (DeiT-Small)	0.4626	**0.7263**

A lower deletion AUC indicates that the regions identified as important are indeed critical to the model's prediction, whereas a higher insertion AUC indicates that the highlighted regions are sufficient to recover the model's confidence. The two methods split the faithfulness comparison: Grad-CAM is more faithful under deletion (confidence collapses faster once its tightly localized high-activation regions are removed), while Attention Rollout is more faithful under insertion (its more diffuse attention pattern is more effective at partially recovering confidence once even a modest fraction of highlighted patches is revealed). This is consistent with the qualitative pattern in Fig. 5: Grad-CAM's sharper, more concentrated activations make it more brittle when its narrow high-activation region does not fall on the pathology (as in the cataract example), whereas Attention Rollout's broader coverage is less precise but more robust to partial evidence. Per-class breakdown (available in `eval.ipynb`) shows both methods are most faithful on glaucoma (deletion AUC of 0.14 for Grad-CAM and 0.09 for Attention Rollout), where the pathology is concentrated at a single, well-localized structure (the optic disc), and least faithful on cataract, matching the qualitative finding that neither method localizes cataract's diffuse presentation well.

E. Discussion

Taken together, the classification and explainability results indicate that overall predictive accuracy and explanation faithfulness are not the same axis of comparison, and neither architecture dominates on both. DeiT-Small edges out ConvNeXt-Tiny on test macro-F1 by a negligible margin (87.74% vs. 87.63%), while ConvNeXt-Tiny is clearly preferable on diabetic retinopathy recall (99.39% vs. 95.76%) and macro ROC-AUC (98.15% vs. 97.06%) – and its paired explanation method, Grad-CAM, is also the more faithful of the two under deletion. For a clinical screening system where missing a diabetic retinopathy case is the costlier error and clinicians need tightly localized visual evidence to trust a prediction, ConvNeXt-Tiny with Grad-CAM is the more defensible choice despite its very slightly lower macro-F1. Conversely, DeiT-Small with Attention Rollout attains marginally higher overall macro-F1 and a higher insertion AUC, which may suit contexts that value broader contextual corroboration over pinpoint localization. This reinforces the motivation stated in Section I: accuracy comparison alone is insufficient for selecting a deployable architecture, and faithfulness metrics surface trade-offs that accuracy hides.

This study has several limitations. The hyperparameter search covered only 3 learning rates $\times$ 5 epoch budgets (15 configurations) per architecture, so the reported results reflect the best configuration found within this grid rather than a globally optimal one. Faithfulness scores were computed on a 40-image subsample of the test set per architecture rather than the full test set, for computational tractability; the resulting deletion/insertion AUC values should be read as estimates. The qualitative comparison in Fig. 5 is illustrative (one image per class) rather than an exhaustive audit, and a

larger-scale clinical validation with ophthalmologist-annotated lesion masks would be needed to more rigorously confirm the anatomical relevance of the highlighted regions. Finally, both models were fine-tuned and evaluated on a single dataset of 4,217 images from one source; generalization to other fundus imaging devices, populations, or class distributions remains untested.

V. Conclusion

Placeholder conclusion text.

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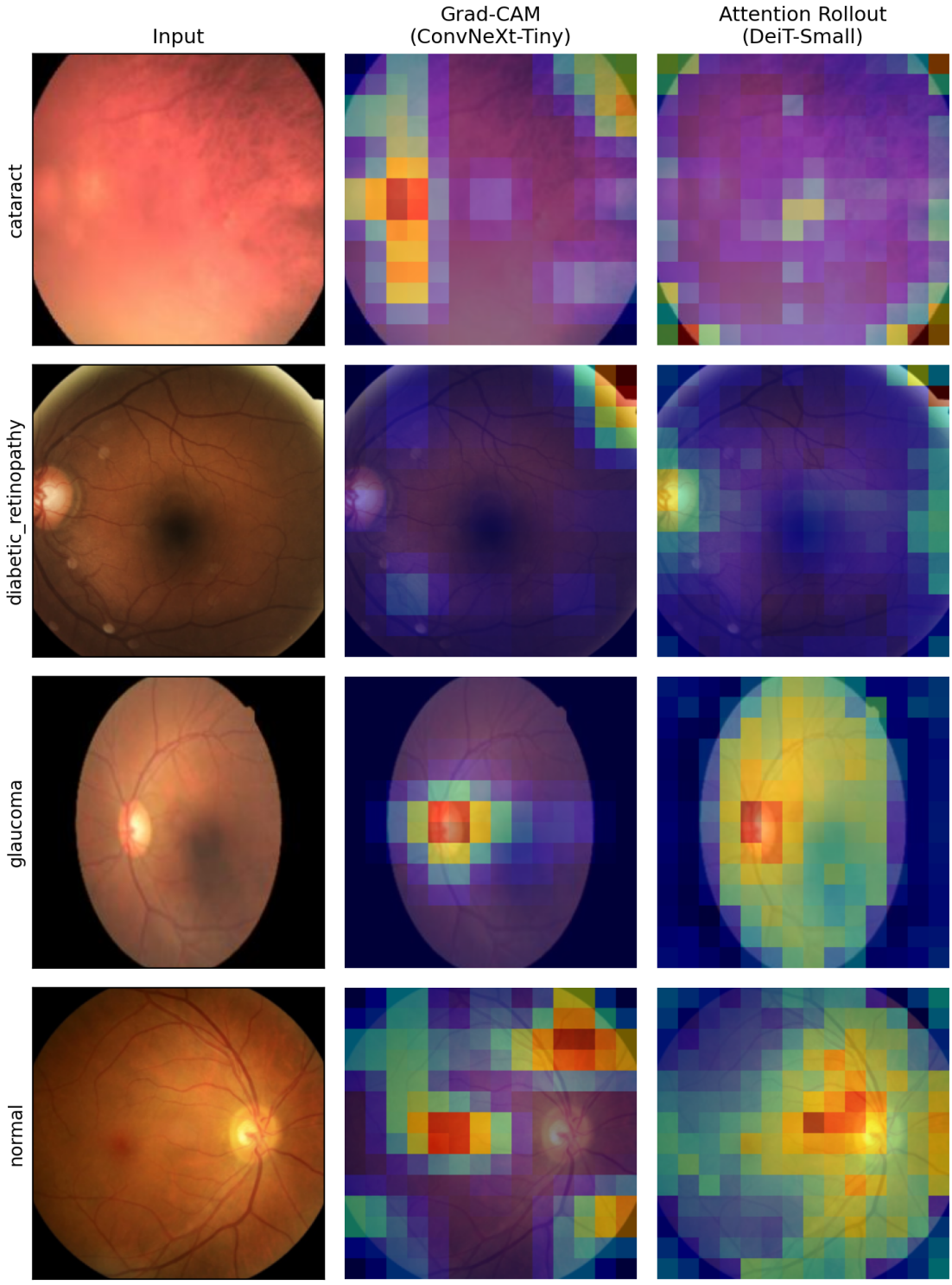


Fig. 5. Qualitative comparison of Grad-CAM (ConvNeXt-Tiny) and Attention Rollout (DeiT-Small) heatmaps across the four eye disease classes.